

Assignment # 1

CPU Scheduling Visualizer

Due Date: November 30, 2025

Details

Students are asked to implement a simple CPU scheduler that takes a list of processes with their burst times (and optionally arrival times), and then shows how the CPU executes them.

The program should:

1. Take user input for:
 - Process ID (e.g., P1, P2, P3)
 - Burst time (in milliseconds or time units)
 - (Optional) Arrival time
2. Implement these scheduling algorithms:
 - FCFS (First Come First Serve)
 - SJF (Shortest Job First)
 - Round Robin
3. Print results showing:
 - Order of process execution (like a simple Gantt chart or list)
 - Waiting time and turnaround time for each process
 - Average waiting time

Deliverables

Submit the following:

1. `source` — Actual code in Python/C/Java
2. Screenshot of the execution of the program capture
3. A `report.pdf` file containing all the steps you have performed and an explanation of the code.